

Hands on experience Training "Python & Machine Learning & AI" Entry Level Training

مدخل للتدريب التقني لتعلم الذكاء الاصطناعي ولغة الالة (Python - Machine Learning - AI)

Python programming for beginners

Duration: 20 hour

Objectives:

- To learn how to design and program Python applications.
- To learn how to use lists, tuples, and dictionaries in Python programs.
- To learn how to identify Python object types.
- To learn how to use indexing and slicing to access data in Python programs.
- To define the structure and components of a Python program.
- To learn how to write loops and decision statements in Python.
- To learn how to write functions and pass arguments in Python.
- To learn how to read and write files in Python.
- To learn how to design object-oriented programs with Python classes.
- To learn how to use class inheritance in Python for reusability.

Course Outline:

1- Python Basics

- Running Python Hello, World!

- Literals
- Python Comments Data Types and Variables
- Writing a Python Module
- “print()” Function
- Named Arguments
- Collecting User Input
- Getting Help

2- Functions and Modules

- Defining Functions
- Variable Scope
- Global Variables
- Function Parameters
- Returning Values
- Importing Modules

3- Math

- Arithmetic Operators
- Modulus and Floor Division
- Assignment Operators
- Built-in Math Functions
- The math Module
- The random Module
- Seeding

4- Python Strings

- Quotation Marks and Special Characters
- String Indexing
- Slicing Strings

- Concatenation and Repetition
- Common String Methods
- String Formatting
- Built-in String Functions

5- Iterables: Sequences, Dictionaries and Sets

- Definitions
- Sequences
- Unpacking Sequences
- Dictionaries
- The len() Function
- Sets

6- Flow Control

- Conditional Statements
- The “is” and “is not” Operators
- Python's Ternary Operator
- Loops in Python
- The enumerate() Function
- Generators
- List Comprehensions

7- File Processing

- Opening Files
- The “os” and “os.path” Modules

8- Exception Handling

- Wildcard except Clauses
- Getting Information on Exceptions
- The else Clause
- The finally Clause
- Using Exceptions for Flow Control
- Exception Hierarchy

9- Dates and Times

- Understanding Time
- The time Module
- The datetime Module

10- Running Python Scripts from the Command Line

11- Classes and Objects

- New Style Classes
- Inheritance
- Properties and Slots
- Static and Class Methods
- Abstract Base Classes
- Method Overriding

Machine learning

Duration: 40 hour

Course Outline:

1: Welcome to the course

- Application of ML
- Install anaconda
- Jupyter Notebook

2: Python Packages (Numpy and Pandas)

- Numpy Array
- Numpy Array Indexing
- Numpy Operation
- Pandas Series
- Pandas DataFrame
- Pandas Missing Data
- Pandas Group-by
- Pandas Merging Joining and Con
- Pandas Operations
- Pandas Data Input and Output

3: Data Reprocessing

- How to get the Dataset
- How to Import The Libraries and Dataset
- Missing Data
- Categorical Data
- Splitting the Data
- Feature Scaling
- Preprocessing Data Template

4: Simple Linear Regression

- Business Problem Description
- Simple Linear Regression Intuition
- Simple Linear Regression in Python
- Dataset + Business Problem Description

5: Multiple Linear Regression Intuition

- Multiple Linear Regression Intuition
- Multiple Linear Regression in Python

6: Polynomial Regression

- Polynomial Regression Intuition
- Polynomial Regression in Python
- Polynomial Regression Template

7: Support Vector Regression (SVR)

- SVR Intuition
- SVR in Python

8: Decision Tree Regression

- : Decision Tree Regression
- : Decision Tree Regression in Python

9: Random Forest Regression

- Random Forest Regression Intuition
- Random Forest Regression in Python

